MILITARY SPECIFICATION

MOTOR, BRAKE ASSEMBLY: 11747088

This specification is approved for use by the U.S. Army Armament Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements, quality assurance provisions, and packaging for a Motor Brake Assembly which is a subassembly of the Sight Current Generator Test Set 11747100 which forms a part of the Electronics System Test Set type AN/TSMIOOA.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-T-5422	-	Testing, Environmental, Aircraft Electronic Equipment
MIL-F-13926	-	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-P-14232	.	Parts, Equipment and Tools for Army Materiel; Packaging and Packing of

FSC 1285

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be *of* use in improving this document should be addressed to: Commander, USArmy Armament Research and Development Center, Attn. DRSMC-QA, Dover,New Jemey07801by using theself-addressed Standardization Document Improvement Proposal (DDForm 1426) appearing at the end of this document or by letter.

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MIL-I-45607 – Inspection Equipment, Supply and Maintenance of
MIL-C-60898 – Circuit Card Assemblies, General Specification for

STANDARDS

MILITARY

MIL-STD-105 – Sampling procedures and Tables for Inspection by Attributes
MIL-STD-109 – Quality Assurance Terms and Definitions
MIL-STD-454 – Standards General Requirements for Electronic Equipment
MIL-STD-810 – Environmental Test Methods

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications-form a part of this specification to the extent specified herein.

DRAWINGS (See 6.4)

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT CENTER (ARDC)

11747088 Motor, Brake Assembly

PACKAGING DATA SHEET

11747088 Packaging of Motor, Brake Assembly

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer).

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3* REQUIREMENTS

3.1 Fabrication. The motor, brake assembly shall be manufactured in accordance with drawing 11747088 and drawings pertaining thereto.

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3.2 General specification. Where applicable, the assemblies shall meet the requirements of MIL-F-13926, as follows:

a. Order of precedence.

b. Dimensions and tolerances.

c. Inorganic protective surface finishes.

d. Part identification and marking.

e. Workmanship.

f. Soldering electrical connections.

3.3 Environmental service conditions.

3.3.1 Vibration. The Motor, Brake Assembly shall be mounted on a vibration table in normal service position. The Motor, Brake Assembly shall meet the requirements of 3.4 after being subjected to a simple harmonic motion having excursions as follows:

Frequency	Excursion (Double Amplitude)
5 to 15 Hz	0.06 inch
15 to 25 Hz	0.04 inch
25 to 55 Hz	0.02 inch

The frequency shall be varied logarithmically between 5 and 55 Hz within 15 minutes and returned to 5 Hz in 15 minutes. The motion shall be applied to the assembly for a period of 30 minutes in each direction of the 3 mutually perpendicular axes (total of 90 minutes), relative to its normal operating position. Vibration tests shall be conducted under ambient room conditions.

3.3.2 Temperatures.

3.3.2.1 Operating. The Motor, Brake Assembly shall meet the requirements of 3.4 while exposed to temperatures of +32°F and +131°F.

3.3.2.2 Storage. The Motor, Brake Assembly shall meet the requirements of 3.4 at temperatures of 60°F to 90°F after being exposed to each temperature of -80°F and +185°F.

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3.3.3 **Humidity** Motor, Brake Assembly shall meet the requirements of 3.4 after being subjected to ten consecutive humidity cycles at relative humidities between 85 and 95 percent and temperatures from 82°F to 160°F over a period not less than 240 hours.

3.3.4 Shock. The Motor, Brake Assembly shall meet the requirements of 3.4 after being subjected to a total of 18 shocks consisting of three shocks in opposing directions along each of three mutually perpendicular axes. Each shock shall have an acceleration of 15 gravity units (G's) and a pulse duration of 11+ 1 millisecond.

3.3.5 Altitude.

3.3.5.1 Operating The Motor, Brake Assembly shall meet the requirements of 3.4 during and after exposure to barometric pressure encountered at 10,000 feet above sea level (20.6 inches Hg).

3.3.5.2 Nonoperating. The Motor, Brake Assembly shall meet the requirements of 3.4 after exposure for not less than 1 hour to barometric pressures encountered at 40, 000 feet above sea level (5.5 inches Hg).

3.3.6 Sand and dust. The Motor, Brake Assembly shall meet the requirements of 3.4 in both operating and nonoperating condition, during and after exposure to sand and dust particles as encountered in desert area.

3.3.7 Salt spray. Brake Assembly shall meet the requirements of 3.4 in both operating and nonoperating condition, during and after exposure to a salt-sea atmosphere.

3.3.8 Fungus-inert materials. The requirements for fungus-inert materials shall be in accordance with MIL-STD-454, Requirement 4.

3.4 Performance.

3.4.1 Functional.

3.4.1.1 Conditions of operation. Prior to functional test, establish and mark the polarity of the brake assembly. The motor and brake input power requirements and the motor output requirements under the various brake load conditions shall be as specified in Table 1.

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TABLE I

Mot or	Input power	Brake	Motor output Speed in RPMs
	115V RMS + 5% Line to line, 3 phase, 400 Hz ± 1%, Total Harmonic Waveform Distortion <, 5% of Voltage of Fundamental Frequency, maximum difference between 3 phase voltages ≤ 2 volts, displacement between adjacent phases shall be within limits of 120° ± 1.5°.	None	11,910 to 11,970(1)
	Same as above.	24.0+0.5 VDC In series with 2370hm + 1% resistor	11,600 to 11,850
	Same as above.	24.0+0.5 VDC in series with 158ohm ± 1% resistor	11,200 to 11,600

(1) Motor shall attain this no-load speed within a maximum of 15 seconds.

3.5 First article inspection. This specification makes provisions for First Article Inspection. Requirements for the submission of First Article Samples by the contractor shall be as specified in the contract.

3.6 Workmanship. The requirements of MIL-STD-454 and other applicable specifications shall apply.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements, as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 General provisions. The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Definition of inspection terms shall be as listed in MIL-STD-109.

4.2 First article (initial production) approval. The requirements for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests shall be 3 test sets manufactured in the same manner using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.2.1 Government testing. When the Government is responsible for conducting article-approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to insure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for materials.

4.2.2 Contractor testing. When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certificates of conformance for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection.

4.3 Inspection provisions.

4.3.1 Submission of product. Unless otherwise specified herein, or by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Products" as specified in MIL-STD-105.

4.3.2 Examinations and tests. Except where otherwise specified herein, components and subassemblies shall be inspected in accordance with the "Inspection provisions for components and subassemblies" and the Sampling Plan Tables contained in MIL-F-13926. Examinations and tests related to Section 3 herein shall be performed in accordance with Tables II and III and test methods and procedures in 4.5. Examinations and tests for packaging, packing and marking shall be in accordance with MIL-P-14232 and Section 5 herein.

4.3.2.1 Components and subassemblies. Printed circuit subassemblies shall inspected in accordance with the inspection provisions of MIL-C-60898. All other components and subassemblies shall be inspected to the applicable requirements of MIL-F-13926.

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4.3.2.2 **Control tests.** One test set shall be selected at random as a control sample from each 25 produced. The control samples shall meet the requirements of Table III after they have met the requirements and tests of Table II.

Table II. Special sampling

Test	Requirement	Test procedure
Fabrication	3.1	Applicable Drawings - Visual
General Specifications	3.2	Visual - tactile
Vibration	3.3.1	4.5.1.1, 4.5.1.2
Temperature	3.3.2	4.5.1.1, 4.5.1.3
Humidity	3.3.3	4.5.1.1, 4.5.1.4
Shock	3.3.4	4.5.1.1, 4.5.1.5
Altitude	3.3.5	4.5.1.1, 4.5.1.6
Sand and Dust	3.3.6	4.5.1.1, 4.5.1.7
Salt Spray	3.3.7	4.5.1.1, 4.5.1.8
Fungus	3.3.8	4.5.1.1, 4.5.1.9

4.3.3 **Acceptance and rejection.** Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.3.4 Functional tests. The requirements and tests in Table III shall inspected *on* a 100% basis. Unless otherwise specified, all examinations and tests shall be conducted at the following standard ambient conditions:

a. Temperature +50° to +90°F

b. Relative humidity 90% RH max.

c. Pressure 28 to 32 inches of Hg

Table 111. Classification of defects
100% inspection

Characteristic	Requirement	Procedure
101. Functional	3.4.1	4.5.2
102. Conditions of operation	3.4.1.1	4.5.2.1
103. No-Load speed	3.4.1.1	4.5.2.2
104. Brake load speeds	3.4.1.1	4.5.2.3
105. Motor direction reversal	3.4.1.1	4.5.2.4

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4.4 Inspection equipment. Except as otherwise provided for by the contract, the contractor shall supply equipment in accordance with the applicable requirements of MIL-I-45607. The government reserves the right to use the test equipment for its own independent inspections to the extent that such use will not unduly interfere with the contractor's delivery schedule.

4.4.1 Contractor design. The contractor shall provide inspection equipment with the "Methods of inspection" specified in 4.5 and with the "Component Inspection" and "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit and permit positive rejection when nonconforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.4.1.1 Test-equipment. In conjunction with 4.4.1, the following test equipment shall be utilized in the performance of the applicable tests as specified in 4.5:

Item		Description
1.	Power Supply	115 VAC ± 5% RMS, 3 phase, 400 Hz ± 1%, total harmonic waveform distortion ± 5% of Voltage of Fundamental Frequency, maximum difference between 3 phase voltages < 2 volts, displacement between adjacent phases shall be within limits of 120° + 1.5°.
2.	Power Supply	24.0 ± 0.5 VDC, 5 amps max., load regulation 1% or better from 0 to 5 amps, less than 1% ripple.
3.	Switch S1	Single pole, 3 position, 5 amp
4*	Switch S2	Single pole, 2 position, 5 amp
5.	Relay	3 PST, NO, 5 amp
6.	Resistor	237 ohms, 1%, 2.5 watts

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7.	Resistor	158 ohms, 1%, 4 watts
8.	Tachometer	O to 13,000 RPM, ± .5% accuracy
9.	Ammeter	O to 5 amp ± 1% accuracy

4.5 Methods of Inspection

4.5.1 Environmental.

4.5.1.1 Test conditions. The test set shall be mounted in an upright position a manner similar to that in which it will be in service and shall be subjected to the environmental tests of the following subparagraphs. The atmospheric pressure shall be at room ambient. Except for the high and low temperature tests or the humidity tests, as applicable the temperature and humidity shall be at room ambient.

4.5.1.2 Vibration. The vibration test shall be made on a device with means provided for measuring and controlling the direction of vibration and adjusting the amplitudes and frequencies to keep them within the specified limits of 3.3.1. The measuring instruments of the testing device shall be capable of measuring the vibration amplitude within plus or minus 5 percent, and capable of measuring the frequency within plus or minus 2 percent. The test shall be performed in accordance with the requirements specified in 3.3.1. Subsequent to the vibration test, the test set shall show no damage or defect resulting from the test when examined by visual and tactile means. The test set shall then be tested to determine compliance with requirements specified under 3.3.1.

4.5.1.3 Temperature. Test in accordance with Procedure I of Method 501 in MIL-STD-810 for high temperature operating and non-operating under conditions specified in 3.3.2. Test in accordance with Procedure I of Method 502 for low temperature operating and nonoperating conditions specified in 3.3.2, and verify requirements of 3.3.2.1 and 3.3.2.2.

4.5.1.4 Humidity. The humidity test shall be performed in accordance with Procedure I of Method 502 in MIL-STD-810. Subsequent to the humidity test, the test set shall be tested to determine compliance with 3.3.3.

4.5.1.5 Shock. The shock test shall be performed in accordance with MIL-T-5422, Part I.

4.5.1.6 Altitude. The altitude test shall be performed in accordance with the following steps:

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Step 1. Place the test set in the test chamber in a manner that will simulate service usage, making connections and attaching instrumentation as necessary. Plugs, covers, and inspection plates not used in operation, but used in servicing shall remain in place. When mechanical or electrical connections are not used, the connections normally protected in service shall be adequately covered. For tests where temperature values are controlled, the test chamber shall be at standard ambient conditions when the test set is installed. The test set shall then be operated to determine that no malfunction or damage was caused due to faulty installation or handling. The requirement for operation following installation of the test set in the test facility is applicable only when operation is required during or immediately following exposure to the specified test. Standard ambient temperature shall be maintained during the entire test.

Step 2. Reduce the chamber internal pressure to 87.5 mm of Hg (3.4 in. of Hg or 50,000 ft. above sea level). The rate of pressure change may be the maximum attainable by the chamber. Maintain the chamber pressure for a period of not less than one hour.

Step 3. Increase the chamber pressure to 523 mm of Hg (20.6 in. of Hg or 10,000 ft. above sea level) and then operate the test set and compare the results with the data obtained in accordance with pretest performance record made of all data necessary to determine compliance with required performance.

Step 4. With the test set not operating, return the chamber to standard ambient conditions.

Step 5. Operate the test set and compare the data obtained with the data obtained in accordance with pretest performance record.

Step 6. The test set shall be visually inspected and a record made of any damage or deterioration resulting from the test. If a test chamber is used for the test, perform a visual inspection of the test within the chamber at test conditions, when possible. Upon completion of the test, visually inspect the test set again after the test set has been returned to standard ambient conditions. Deterioration, corrosion or change in tolerance limits of any internal or external parts which in any manner prevent the test set from meeting operational service or maintenance requirements shall provide reason to consider the test set as having failed to withstand the conditions of the test.

4.5.1.7 Sand and dust. The sand and dust test shall be performed in accordance with MIL-T-5422.

4.5.1.8 Salt spray. The salt spray test shall be performed in accordance with MIL-T-5422.

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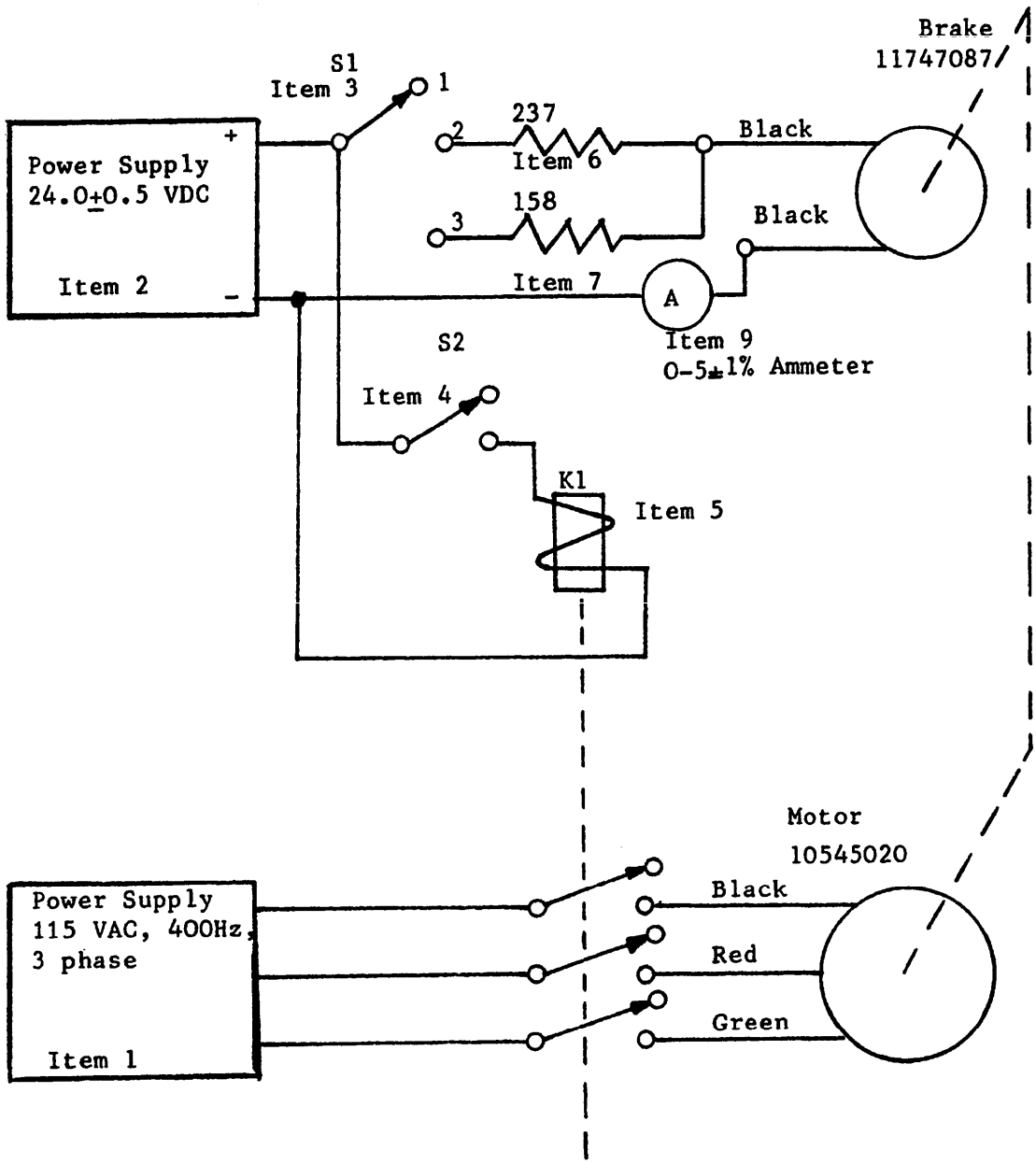


Figure 1. Motor, Brake test setup

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4.5.1.9 Fungus. The fungus resistance test shall be performed in accordance with MIL-T-5422.

4.5.2 Functional tests.

4.5.2.1 Conditions of operation. Connect test equipment, paragraph 4.4.1.1 and motor brake assembly as shown in figure 1. With item 3 (S1) to pos. 3 and S2 closed, verify positive (+) polarity using ammeter, item 9, paragraph 4.4.1..1. To designate a positive (+) polarity for the brake assembly, a maximum current indication has to be established. This is obtained by interchanging the leads at the brake assembly with item 6 and 9, paragraph 4.4.1.1 as shown in figure 1. Once maximum current is established, note which lead from brake assembly is now in series with item 6 and the positive side of item 2. That Lead is now designated the "positive" side and should be labeled accordingly. Connect test equipment, paragraph 4.4.1.1, and motor brake assembly as shown in figure 1 and verify motor speed using tachometer, item 8, paragraph 4.4.1.1.

4.5.2.2 No-load speed. With conditions per 4.5.2.1, close Switch S2 and motor shall attain, within a maximum of 15 seconds, and maintain a speed of 11,910 to 11,970 RPMs.

4.5.2.3 Brake load speeds. With the motr operating at No-load speed per 4.5.2.2, switch S1 to position 2 and the motor shall decrease in speed to between 11,600 and 11,850 RPMs: A further reduction to between 11,200 and 11,600 RPMs shall be attained when S1 is switched to position 3.

4.5.2.4 Motor direction reversal. Interchange any two motor lead connection and repeat procedures 4.5.2.2 and 4.5.2.3 respectively; motor shall rotate per 4.5.2.2 and 4.5.2.3 but in opposite direction to previous direction.

5* PACKAGING

5.1 Preservation, packaging, packing, and marking. Preservation, packaging, packing, and marking shall be in accordance with Packaging Data Sheet 11747088.

6. NOTES

6.1 Intended *use*. *The* Motor, Brake Assembly is a subassembly of the Sight Current Generator Test Set 11747100 which forms a part of the AN/TSM-100A Electronic Systems Test Set.

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6.2 **Ordering data.** Procurement documents should specify the following;

a. Title, number and date of this specification.

b. Applicable stock number.

c. A block of serial numbers to identify each test set.

d. Instructions for first article inspection.

6.3 **Submission of designs for approval.** Contractor designs shall be approved by the Government prior to fabricating or procuring of the equipment. Partial submission of inspection equipment designs is permissible and encouraged. However, the completion date for design review will be based on the date of the final submission of designs and the required delivery schedule as stipulated in the contract. Submit designs as required to: Commander, US Army Armament Research and Development Center, ATTN : DRSMC-QAF-I(D), Dover, NJ 07801.

When the contractor submits inspection equipment designs to the Government for approval, he shall give the following information in his letter of transmittal:

a. The contract number.

b. The contract item (name, model number, etc.).

cm The designs remaining to be submitted and the expected date of transmittal.

6.4 **Drawings.** Drawings listed in Section 2 of this specification under the heading US Army Armament Research and Development Center (ARDC) may also include drawings prepared by, and identified as Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, ARRADCOM, or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARDC.

6.5 **Changes to previous issue.** Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-M-48598A (AR)	Motor, Brake Assembly: 11747088

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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